

GCC to compile GSL with any problems. You can either update your version of GCC, or alternately, download an old version of GSL. To compile the code, issue the following commands:

```
$ ./configure
$ make
$ make check
```

(The third command above is not necessary, but I recommend it so you can check that all went well with the compilation. Figures 1 to 3 illustrate the compilation process.)

For the final step, you will need root privileges. Use the appropriate mechanism—*sudo* or *su*—to attain root privileges, and run the command given below:

```
# make install
```



Note: A list of known issues that you may encounter while installing GSL is addressed in the file *INSTALL* in the archive-extracted directory.

Our first GSL program

Having installed GSL, we now start with our first GSL program. Those who know how to make and use header files will understand this part easily. For our first program, I have chosen to take an example from complex numbers; as far as I know, handling complex numbers in C/C++ is quite tedious. This program, which we will save as *first_gsl.c*, performs the following operations:

It will take the polar form variables for magnitude (*x*) and theta (*y*) and convert the values into a rectangular complex number (*z*).

To cross-check, it will again find the magnitude (*abs*) of the complex number (*z*).

```
#include <stdio.h>
#include <gsl/gsl_complex.h> /*1. GSL header files */
#include <gsl/gsl_complex_math.h>

int main (void)
{
```

```
File Edit View Terminal Help
~ amol@amol-desktop:~/Desktops$ gcc -Wall -I/usr/local/include -c first_gsl.c
~ amol@amol-desktop:~/Desktops$ gcc -I/usr/local/lib first_gsl.o -lgsl -lgslcblas -lm
~ amol@amol-desktop:~/Desktops$ ./a.out
The complex number is:
3.535178440839203940 + i 3.535893934360332391
The magnitude of the complex number is:
5.00000000000000000000000000000000
~ amol@amol-desktop:~/Desktops$ gcc wall first_gsl.c -lgsl -lgslcblas -lm -o first_gsl
~ amol@amol-desktop:~/Desktops$ ./first_gsl
The complex number is:
3.535178440839203940 + i 3.535893934360332391
The magnitude of the complex number is:
5.00000000000000000000000000000000
~ amol@amol-desktop:~/Desktops$ gcc -Wall -c first_gsl.c
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~ amol@amol-desktop:~/Desktops$ ./a.out
The complex number is:
3.535178440839203940 + i 3.535893934360332391
The magnitude of the complex number is:
5.00000000000000000000000000000000
~ amol@amol-desktop:~/Desktops$
```

Figure 4: Compiling and running the program

```
double x = 5.0;
double y = 0.7855;
gsl_complex z; /*2. GSL data structure */
z = gsl_complex_polar (x, y ); /*3. GSL function 1*/
printf("The complex number is:\n");
printf("%0.18f + i %0.18f\n",z.dat[0],z.dat[1]); /*4. printing return values*/

double abs;
abs=gsl_complex_abs(z); /*Another GSL function*/
printf("The magnitude of the complex number is:\n ");
printf("%0.18f\n",abs);
return 0;
}
```

You can compile this program in two ways—with linking as a separate step, or combined with the compilation. First, we look at the version using separate statements to compile, link, and run the output program:

```
$ gcc -Wall -I/usr/local/include -c first_gsl.c
$ gcc -I/usr/local/lib first_gsl.o -lgsl -lgslcblas -lm
$ ./a.out
```

We can compile and link in a single step, and also provide a nice name for the compiled binary:

```
$ gcc -Wall first_gsl.c -lgsl -lgslcblas -lm -o first_gsl
$ ./first_gsl
```

The output produced by the program is shown in Figure 4.

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